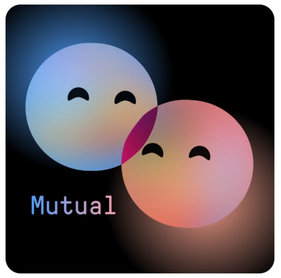
CAL HACKS 13.0 · HCI · UC BERKELEY, OCTOBER 2026

Human Human Interaction Lab

Filtering the crowd so every hello counts: a reciprocal, proximity aware name badge on the phone you already own

Ethan Pan · Paul Jiang · Jiayi Tang · Kenny Nguyen

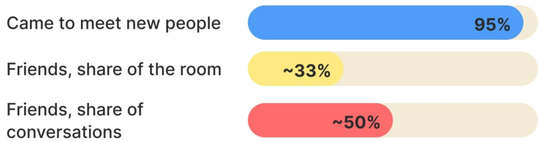
UC Berkeley · UC Irvine · Simon Fraser University



1 The problem

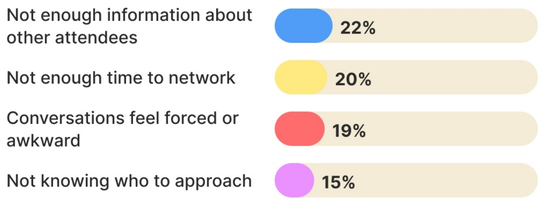
Hackathons, conferences, career fairs, orientation: you get a few hours to find a team, a cofounder or a mentor among hundreds of strangers. Name tags say nothing about fit, so every conversation is a gamble with your limited time, and the right person may be ten feet away.

People don't mix at mixers



Ingram and Morris, Administrative Science Quarterly, 2007. ~100 executives wearing electronic name tags at a mixer.
Source: doi.org/10.2189/asqu.52.4.558

Why networking at events falls short



Each attendee named their single biggest challenge; these four add up to 76%. Bizzabo 2025 Event Networking Report, survey of 300+ event organizers, attendees and sponsors. Industry survey.
Source: welcome.bizzabo.com/hubfs/Q225_NetworkingReport.pdf

38% of professionals find it hard to stay in touch with their network. The top reason: not enough time.

People value their contacts, but few keep them up:



LinkedIn global survey of 15,905 members in 17 countries, Feb to Mar 2017 (LinkedIn Pressroom, June 2017). The top reason was cited by 49%. Industry survey.
Source: news.linkedin.com/2017/6/eighty-percent-of-professionals-consider-networking-important-to-career-success

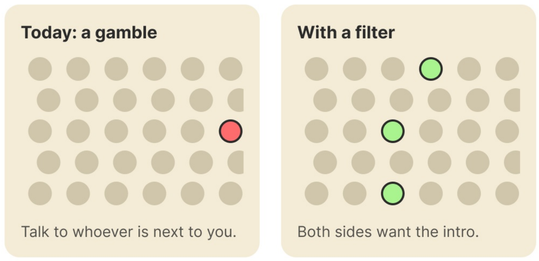
2 Our thinking

THE QUESTION

Can we make each encounter more valuable on average?

Not more conversations. Better odds for each one.

Filtering beats picking at random



Evidence: meeting the right people is the bottleneck. Researchers randomly placed in one shared 90 minute session were 75% more likely to apply for a grant together (Boudreau et al., Review of Economics and Statistics 2017). When a hurricane cancelled APSA 2012, would be attendees became 16% less likely to coauthor (Campos et al., Economic Journal 2018).
Sources: doi.org/10.1162/REST_a_00676 · doi.org/10.1111/ecoj.12566

What we believe

- **Filter before you talk**
  Reciprocal: an intro only happens when both people would want it.
- **Keep eyes up**
  The signal is worn on your chest, not buried in another feed.
- **Lower the fear**
  A shared color says: the other person wants this too.
- **Close the loop**
  A summary and a follow up arrive by text after you talk.
- **No new hardware**
  The phone you own, in a 3D printed lanyard holder.


Cal Hacks 13.0: hundreds of hackers, one hangar

3 Related work

MIT Media Lab badges

Borovoy et al. 1998 · Paradiso et al. 2010

Did: Meme Tags swapped short ideas face to face; later badges sensed who met whom.
Inspired: give strangers something to talk about.
Different: no custom hardware; topics flip to what this person wants to discuss with you.


Our drawing · Meme Tag, 1998 · UbER-Badge, 2010 · Communicator badge

Photos from the papers: Borovoy et al., CSCW 1998, Fig. 1; Paradiso et al., Personal and Ubiquitous Computing 2010, Figs. 1 and 18 (MIT Media Lab).

Social Serendipity

Eagle and Pentland, published in IEEE Pervasive Computing 2005

Did: Bluetooth phones alerted two nearby people whose profiles matched.
Inspired: an intro needs both sides to clear the bar.
Different: an LLM scores needs in plain language, and the cue is worn, not a phone alert.


Figure from Eagle and Pentland, IEEE Pervasive Computing 2005

SocialStools

Guo, Leshed and Green, published at CHI 2023

Did: stools whose distance drives shared sound and light between strangers.
Inspired: map physical distance to a shared cue.
Different: our cue is a color both badges take on, only for mutual matches.


Our drawing of distance as a shared cue

RECON

Pizzato et al., published at RecSys 2010

Did: a reciprocal recommender for online dating.
Inspired: score each direction of interest.
Different: a geometric mean of LLM fit, used in the physical room.

What's new here

- Matching that needs both sides, scored by an LLM from natural language
- A physical, ambient signal you wear, not a phone alert
- Peer to peer UWB: no venue readers, no custom badge
- The loop closes after the chat: summary, follow up, a personal CRM

Next: measure intros per hour, and how many saved contacts become a real follow up.

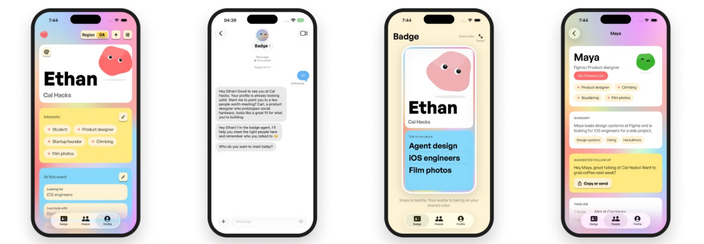
4 Our solution

A live badge you wear on a lanyard



Worn on a lanyard (AI rendering) · Alone: your name and topics · Match nearby: shared tint · Talking: avatars hop in

How it works



1 Onboard — Text Photon what you want and offer. Claude builds your profile.

2 Heads up — Pairs are scored both ways. Only mutual fits get a text.

3 Walk closer — UWB distance tints both avatars toward one shared color.

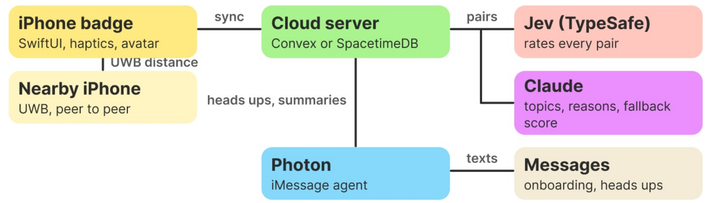
4 Follow up — Then a summary and follow up text. People keeps who you met.

RECIPROCAL SCORE: JEV RATES EVERY PAIR, CLAUDE WRITES WHY

$$\text{score} = 0.75\sqrt{\text{fit}_{\text{me}} \cdot \text{fit}_{\text{them}}} + 0.25\,\text{values}$$

Jev (TypeSafe) answers three rubric questions per pair (fit for me, fit for them, will we click) in about half a second. Claude writes the badge topics and reasons, and scores if Jev is unavailable. A geometric mean punishes one sided interest: fits of 0.9 and 0.1 score 0.30.

Architecture



References: Ingram & Morris, Administrative Science Quarterly 2007 · Bizzabo, 2025 Event Networking Report · LinkedIn global networking survey, 2017 · Boudreau et al., Review of Economics and Statistics 2017 · Campos, Leon & McQuillin, Economic Journal 2018 · Borovoy et al., CSCW 1998 · Paradiso et al., Personal and Ubiquitous Computing 2010 · Eagle & Pentland, IEEE Pervasive Computing 2005 · Guo, Leshed & Green, CHI 2023 · Pizzato et al., RecSys 2010. TypeSafe Jev, docs.typesafe.ai. Worn badge photo is an AI rendering made with ChatGPT.

Built at Cal Hacks 13.0 with Convex, Claude, Photon and Figma.
Demo and Devpost: devpost.com/software/human-human-interaction-lab